

Microbiological Methods

AMR Threat Perception Assessment of Heterotrophic Bacteria From Shrimp Aquaculture Through Epidemiological Cut off Values

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Abstract

Background: Emergence and dissemination of antibiotic resistance is one of the major risks associated with the rampant usage of antibiotics in food-producing animals including aquaculture.

Objective: To determine Epidemiological Cut-OFF (ECOFF) values of heterotrophic bacterial populations from shrimp culture environments against five different antibiotics

Methods: In this present study, bacterial samples were isolated from *Penaeus vannamei* culture environment in different locations of Andhra Pradesh, which is the aquaculture hub of India. The bacterial isolates were assessed for antibiotic resistance towards five antibiotics belonging to different classes (oxytetracycline, chloramphenicol, erythromycin, ciprofloxacin, and co-trimoxazole) by the disc diffusion method. Determination of Epidemiological Cut-OFF (ECOFF) values and analysis by employing normalized resistance interpretation (NRI) was carried out.

Results: The most dominant bacterial populations from shrimp culture were *Vibrio* spp. (pathogenic bacteria) followed by *Bacillus* spp. (probiotic bacteria). The bacterial isolates showed highest resistance towards oxytetracycline (overall 23.38%) and in location L6 (59.4%) followed by co-trimoxazole (31.1%). ECOFF values calculated by employing NRI showed that the disc diffusion data were distributed in a normalized manner. The maximum ECOFF value was obtained for ciprofloxacin (23.32 mm), while the minimum value was observed for oxytetracycline (9.05 mm). The antibiotic resistant phenotypes showed that the majority of the heterotrophic bacterial isolates (>60%) belonged to the non-wild type phenotype and primarily towards oxytetracycline (90%).

Conclusion: The presence of non-wild antibiotic-resistant phenotypes of heterotrophic bacterial populations (which include not only pathogenic bacteria but also probiotic bacteria) indicates that shrimp culture ponds may be a reservoir for drug-resistant bacteria and there is a greater risk associated with transmission of resistant genes across bacterial flora.

Highlights: NRI analysis of antibiotic disc diffusion data of heterotrophic bacterial populations in shrimp aquaculture environments revealed that majority of them belonged to non-wild type (90%) particularly to oxytetracycline in comparison to other studied antibiotics (chloramphenicol, erythromycin, ciprofloxacin and co-trimoxazole).

White leg shrimp (*Penaeus vannamei*) is a widely cultured crustacean species contributing significantly to global aquaculture production (5.8 million tonnes; 1). Farmed *P. vannamei* production in India is 815 745 tonnes, with the major contribution from the state of Andhra Pradesh (639 896 tonnes; 2). Antibiotics have been applied in animal production systems including shrimp aquaculture, either for growth promotion or disease management (3). Tetracycline, ciprofloxacin, erythromycin, chloramphenicol, and co-trimoxazole are the antibiotics generally used

in the world for aquaculture practices (4). Excess and frequent use of antibiotics in animal rearing systems has resulted in detrimental health issues in humans and become a great concern in the development and acquisition of bacterial resistance towards antibiotics (5).

Several authors have reported the presence of antibiotic, as well as multidrug, resistance in bacteria from shrimp culture systems in recent years (6–9). Hence, aquatic ecosystems have been regarded as important sources of drug-resistant microorganisms

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and could be treated as emerging antibiotic pollution (10). These studies have been carried out on antimicrobial susceptibility patterns with regard to a particular pathogenic bacteria, but this strategy will not give the total antibiotic resistance pattern of the culturable bacteria prevailing in the pond ecosystem. In tropical countries the conditions favor the multiplication of heterotrophs as they prevail in higher densities than pathogenic bacteria and the isolation is easy requiring no specific enrichment. In surveillance studies, heterotrophic bacteria will be useful as indicator organisms for defining antibiotic resistance patterns (11). Hence, data on the antimicrobial susceptibility of heterotrophs play a significant role in understanding the spread of resistant genes to other commensal bacteria through mobile genetic elements by means of horizontal gene transfer (HGT; 12).

For studying whether a particular bacterium is susceptible or resistant to an antibiotic, the drug resistance pattern needs to be determined in terms of minimum inhibitory concentration (MIC) values and specific zone of inhibition (zone diameter in mm) and interpreted based on the break-even points as per the CLSI (Clinical and Laboratory Standards Institute) or epidemiological cutoff values of EUCAST (European Committee on Antimicrobial Susceptibility Testing). Understanding the antibiotic resistance pattern and its mechanism is pertinent for disease treatment (13). The concept of Epidemiological Cut-OFF value (ECOFF) was introduced by EUCAST to categorize the bacterial species in terms of antibiotic susceptibility either as wild type (WT) phenotypes or as non-wild-type (NWT) phenotypes (14). The zone of inhibition or MIC for a particular bacterial isolate is equal to or less than ECOFF value are considered as WT phenotypes and if it is greater than the ECOFF value are considered as NWT phenotypes. Normalized resistance interpretation (NRI) has been proposed to analyze the data on the antibiotic susceptibility of bacteria based on the disc diffusion method (15).

Few studies have reported ECOFF values for pathogenic aquatic bacteria in terms of antibiotic resistance and defined cut-off wild type (CO_{WT}) values (16, 17). Recently, ECOFF values for the aquatic pathogen *Aeromonas* against eight antibiotics were determined using MIC data (18). Similarly, ECOFF values for the aquatic pathogen *Streptococcus parauberis*, isolated from aquaculture farms in Korea, were calculated by employing MIC data and NRI data interpretation using ECOFFinder tools (19). These reports define CO_{WT} values specifically for an individual pathogenic bacteria for a particular antibiotic. However, there is a paucity of information on CO_{WT} values for heterotrophic bacteria towards commonly used antibiotics in aquaculture. Hence, this study was undertaken to determine CO_{WT} values of antibiotic-resistant heterotrophic bacteria isolated from shrimp culture environments towards five antibiotics: oxytetracycline, ciprofloxacin, chloramphenicol, erythromycin, and co-trimoxazole.

Experimental Collection of Samples

Semi-intensive shrimp ponds were selected based on area of cultivation (1 to 2 acre) from four districts of the state of Andhra Pradesh, India as the stocking density and feeding practices differ from place to place (Table 1). The shrimp pond samples ($n = 192$) were collected from 12 geographical locations with five ponds from each location that are spread over a distance of 2–6 km from each other. The samples comprised pond sediment ($n = 60$), pond water ($n = 60$), cultured shrimp ($n = 60$), and pond source water ($n = 12$). All the samples were collected during the standing crop at different time intervals during a 2-year period (2017 to

2018). Immediately after collection all the samples were brought to the laboratory under iced conditions for further microbiological analysis.

Isolation and Identification of Bacteria by Conventional Methods

Heterotrophic bacteria were isolated from 192 shrimp pond samples (sediment, water, and shrimp) by employing serial dilution and the spread plate method employing marine agar as per ISO 4833-2: 2013 with slight modifications (9). All the isolates were purified on marine agar and stored at -78°C in 15% glycerol. The isolates were identified up to genus level by conventional biochemical tests as per Chauhan and Jindal (20). Biochemical tests performed includes Gram staining, motility, catalase, oxidase, Methyl Red-Voges Proskauer (MR-VP) test, nitrate reduction, citrate utilization, hydrolysis of epac, gelatin, esculin, and starch, utilization of different sugars, decarboxylation of amino acids, growth in media with different concentrations of NaCl (0–20%), growth at different temperatures (10 – 60°C) and growth at different pH values (5–10).

Molecular Identification of Bacterial Isolates by PCR

The bacterial isolates were selected randomly from each group and identified molecularly by carrying out 16S rRNA sequencing (21). Bacterial isolates were grown overnight in marine broth (BD, Difco) and pelleted by centrifuging at $4629g$ for 10 min. The supernatant was discarded and the pellet was dissolved in TE buffer (1X, pH 8.0). The heat lysis method was used to extract crude DNA. Using universal primers 27F and 1492R, the 16S rRNA region was amplified. The $25\ \mu\text{L}$ PCR reaction mixture consisted of $12.5\ \mu\text{L}$ 2X OneTaq Quick load PCR master mix (New England Biolabs, Germany), forward and reverse primers of 20 pmol concentration each ($0.5\ \mu\text{L}$), target DNA ($2\ \mu\text{L}$), and distilled water ($9.5\ \mu\text{L}$). The PCR program consisted of initial denaturation at 94°C for 2 min, followed by 35 cycles of denaturation, annealing, and extension at 94°C for 30 s, 50°C for 1 min, and 68°C for 90 s, with a final extension at 68°C for 7 min. The amplicons were examined using agarose gel electrophoresis (2%) and visualized in a gel documentation system (Biorad, Germany). The amplicons were purified using a GenElute Gel extraction kit (Sigma, United States), and the sequencing was contracted out to Scigenome Pvt. Ltd (Cochin, India). The obtained sequences were compared with the library available with the NCBI and deposited in the public domain with accession numbers for each bacterial species.

Antibiotic Susceptibility Studies

The bacterial isolates were then screened for resistance against five antibiotics as per the standard disc diffusion method (22) using antibiotic discs (Himedia, India) with concentrations of chloramphenicol ($30\ \mu\text{g}$), ciprofloxacin ($5\ \mu\text{g}$), co-trimoxazole ($10\ \mu\text{g}$), oxytetracycline ($30\ \mu\text{g}$), and erythromycin ($15\ \mu\text{g}$). Mueller-Hinton agar plates were incubated at $35 \pm 1^{\circ}\text{C}$ for 24 h and the zone of inhibition (mm) was measured for each antibiotic and the results later used for the calculation of ECOFF values.

Determination of ECOFF Values and Analysis by NRI

CO_{WT} values of the heterotrophic bacterial isolates were calculated using as NRI Excel graph sheet (<http://www.bioscand.se/nri/>) (23). Zone-wise inhibition values were entered in the Excel sheet as the frequency shown by the total isolates to that particular antibiotic. Moving averages were calculated by smoothing the data by

Table 1. Samples collected from shrimp ponds and their location

District	Shrimp P = ponds (n = 5)	Geographical location (Lat. & Long.)	Samples collected (n = 192)				Total samples (n)
			Sediment (n)	Water (n)	Shrimp (n)	Source water (n)	
Srikakulam	L1	18.58°N; 84.30°E	5	5	5	1	16
East Godavari	L2	16.92°N; 82.21°E	5	5	5	1	16
East Godavari	L3	16.96°N; 82.18°E	5	5	5	1	16
East Godavari	L4	16.52°N; 81.98°E	5	5	5	1	16
East Godavari	L5	16.73°N; 82.21°E	5	5	5	1	16
Nellore	L6	14.34°N; 80.03°E	5	5	5	1	16
Nellore	L7	14.26°N; 80.11°E	5	5	5	1	16
West Godavari	L8	16.46°N; 81.39°E	5	5	5	1	16
West Godavari	L9	16.47°N; 81.49°E	5	5	5	1	16
Srikakulam	L10	18.18°N; 84.84°E	5	5	5	1	16
East Godavari	L11	16.90°N; 82.16°E	5	5	5	1	16
East Godavari	L12	16.93°N; 82.25°E	5	5	5	1	16

means of rolling data. Generally, 4-point means are used for this purpose. However, as suggested by Kronvall and Smith (15) for the calculation of moving averages for aquatic bacteria, 8-point rolling means were used and putative peak values were identified. Based on this, the overall number of WT observations for the normally distributed data were calculated. The cumulative percentage distribution of the probit values were calculated for the WT observations. The probit values were plotted and the slope and intercept were calculated to give the best fit line. From the slope and intercept values, the mean of WT observations and the standard deviation were calculated. COWT was finally arrived from the mean and standard deviation of the normalized WT data.

Results

A total of 2304 heterotrophic bacteria isolated from different samples (sediment, water, shrimp, and source water) from shrimp aquaculture systems were identified by both conventional and molecular assays. The results of conventional biochemical identification of bacteria isolated from shrimp ponds and the accession number obtained after blasting of sequences is presented in Table 2, and see also Supplemental Tables S1a–S1d. The bacteria were categorized into four main morphotinctorial groups based on Gram staining and morphology. Group-1 consisted of Gram-positive rod-shaped bacteria comprising four genera namely *Bacillus* spp., *Microbacterium* spp., *Bhargavaea* spp., and *Exiguobacterium* spp. The most dominant bacteria in this group was *Bacillus* spp. (n = 456). Group-2 consisted of Gram-negative rod-shaped bacteria from seven genera namely *Vibrio* spp., *Halomonas* spp., *Tenacibaculum* spp., *Pseudoalteromonas* spp., *Escherichia coli*, *Pseudomonas* spp., and *Burkholderia* spp. The most dominant bacteria in this group as well as in this study was *Vibrio* spp. (n = 888). Group-2 consists of bacteria pathogenic to aquatic animals and opportunistic pathogens to humans. Group-3 consisted of Gram-positive cocci-shaped bacteria from four genera namely *Kocuria* spp., *Salinicoccus* spp., *Staphylococcus* spp., and *Brachyбактерium* spp. In this group *Staphylococcus* spp. (n = 118) was the dominant group. Group-4 consisted of Gram-negative coccobacilli-shaped bacteria from the genus *Acinetobacter* spp. (n = 78). Location-wise distribution of bacterial isolates indicated that *Vibrio* spp. Were the dominant species in all the locations followed by *Bacillus* spp.

Based on the results of biochemical tests, the bacterial isolates (one each from each genus) were randomly selected from each of the four groups and molecular confirmation was carried out by 16S rRNA sequencing. The sequences obtained were deposited in the NCBI database with the accession number as mentioned in

Table 2. Analysis of 16S rRNA sequences of the bacterial isolates showed close similarity to different bacterial species belonging to 16 genera namely *Acinetobacter soli*, *Bacillus subtilis*, *Bhargavaea beijingensis*, *Brachyбактерium paraconglomeratum*, *Burkholderia epacian*, *Escherichia coli*, *Exiguobacterium mexicanum*, *Halomonas ventosae*, *Kocuria flava*, *Microbacterium aurum*, *Pseudoalteromonas piscicida*, *Pseudomonas knackmussii*, *Salinicoccus roseus*, *Staphylococcus aureus*, *Tenacibaculum lutimaris*, and *Vibrio parahaemolyticus*.

All the isolates were screened for antimicrobial resistance using the disc diffusion method against five antibiotics and the antibiotic resistance (percentage) shown by bacteria is shown in Table 3. Highest resistance towards chloramphenicol was found in bacterial isolates from locations L6 and L7 (11.1%). Importantly, for ciprofloxacin erythromycin, and co-trimoxazole, the bacterial isolates from location L6 recorded the highest resistances (23.3, 26.1, and 31.1%, respectively). In the case of oxytetracycline, the heterotrophic bacterial isolates from locations L6 and L1 showed resistances of 59.4 and 56.6%, respectively. It is evident from Figure 1 that the bacterial isolates showed resistance towards oxytetracycline (23.38%), followed by erythromycin (12.73%), co-trimoxazole (10.00%), ciprofloxacin (9.58%), and chloramphenicol (6.02%). Multi-drug resistance was noticed in 11.85% of the bacterial isolates from shrimp aquaculture system.

The CO_{WT} values were calculated for all 2304 heterotrophic bacteria isolated from shrimp pond samples after performing NRI analysis using the antibiotic disc diffusion data against the five antibiotics (chloramphenicol, ciprofloxacin, erythromycin, co-trimoxazole, and oxytetracycline). The percentage of WT and NWT phenotypes are shown in Table 4. CO_{WT} for chloramphenicol was found to be ≥19 mm from the frequency data of the antibiotic zone, which categorized heterotrophic isolates as NWT (77%) and WT (23%). For ciprofloxacin disc diffusion data, CO_{WT} was calculated as ≥23 mm, which divided the heterotrophic bacterial isolates as NWT (68%) and WT (32%). In the case of erythromycin, the CO_{WT} was calculated as ≥18, which classified the heterotrophic bacterial isolates as WT (26%) and NWT (74%). The CO_{WT} for co-trimoxazole was found to be ≥22 mm, which divides the 2304 heterotrophic isolates into WT (38%) and NWT (62%). For oxytetracycline, CO_{WT} was found to be ≥9 mm that classified 10% of heterotrophic isolates as WT and 90% as NWT. The normalized curves made from the disc diffusion zone values are depicted in Figure 2(a–e).

Discussion

Antimicrobial resistance (AMR) has been recognized as an emerging threat to public health across the globe in recent times (24).

Table 2. Bacteria isolated from shrimp pond samples in different locations

Bacteria	Sample collection location												Total isolates	Gene Bank accession number	Percent identity
	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12			
Group-1: Gram-positive rods															
Bacillus spp.	35	39	32	28	41	41	43	36	45	41	42	33	456	MW295585	100
Microbacterium spp.	5	5	4	5	5	6	5	2	8	7	7	6	65	MW295542	98.2
Bhargavaea spp.	2	1	3	2	3	0	2	2	1	3	2	3	24	MW295821	100
Exiguobacterium spp.	7	8	7	6	8	3	8	6	8	5	4	7	77	MW295833	97.2
Group-2: Gram-negative rods															
Vibrio spp.	72	77	70	79	68	78	79	69	74	79	77	66	888	MW454377	100
Halomonas spp.	11	12	12	13	13	11	10	12	13	10	12	10	139	MW454086	99.6
Tenacibaculum spp.	10	13	11	16	12	13	12	12	15	12	9	14	149	MW454349	99.1
Pseudoalteromonas spp.	4	3	5	4	4	8	6	5	6	6	3	5	59	MW295588	98.5
Escherichia coli	3	4	3	4	5	3	3	2	4	2	6	3	42	MW295580	100
Pseudomonas spp.	10	9	11	12	8	10	9	10	10	11	12	11	123	MW295832	99.6
Burkholderia spp.	1	1	2	0	1	2	1	0	1	0	2	1	12	MW295834	100
Group-3: Gram-positive cocci															
Kocuria spp.	2	3	2	2	3	0	3	2	3	3	0	2	25	MW295835	99.8
Staphylococcus spp.	9	11	7	12	10	11	11	10	10	10	8	9	118	MW295826	98.5
Salinicoccus spp.	4	5	3	4	2	3	4	3	2	2	2	3	37	MW295555	99.8
Brachybacterium spp.	1	1	1	2	0	1	0	1	1	2	1	1	12	MW295574	100
Group-4: Gram-negative coccobacilli															
Acinetobacter spp.	8	7	5	7	6	5	6	5	6	7	9	8	78	MW295538	99.8
Total	184	199	178	196	189	195	202	177	207	200	196	182	2304		

Table 3. Antibiotic resistance of heterotrophic bacterial isolates from shrimp culture ponds

Location	Total No. of isolates	Resistant bacterial isolates, %				
		Chloramphenicol	Ciprofloxacin	Erythromycin	Co-trimoxazole	Oxytetracycline
L1	192	4.5	17.8	18.9	16.1	56.6
L2	192	6.6	9.4	9.4	8.3	26.6
L3	192	4.4	4.4	11.6	8.3	15
L4	192	4.4	11.1	6.1	9.4	27.2
L5	192	6.6	12.7	8.3	10	25.5
L6	192	11.1	23.3	26.1	31.1	59.4
L7	192	11.1	15.5	20.5	11.8	40
L8	192	1.6	0.5	5.5	5	7.7
L9	192	3.3	1.6	6.6	3.3	7.6
L10	192	10	7.7	7.7	5.5	17.7
L11	192	4.4	8.8	8.8	4.4	12.2
L12	192	1.1	1.1	1.6	3.8	6.1

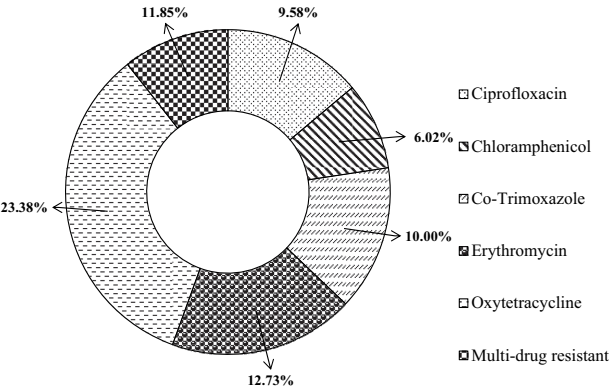


Figure 1. Overall antibiotic resistance percentage of heterotrophic bacteria isolated from shrimp culture environments to five different antibiotics.

Excessive and continuous application of antibiotics in food production systems for therapeutic purposes has resulted in the development of resistance in bacterial populations (18). The World

Table 4. Epidemiological cutoff values (CO_{WT}) values and the frequency of wild-type (WT) and non-wild-type (NWT) types analyzed by NRI from antibiotic disc zone data for heterotrophic bacterial isolates from shrimp ponds

Antibiotic	CO _{WT} , mm	WT, %	NWT, %	Standard deviation, mm
Chloramphenicol	19.48	23	77	4.14
Ciprofloxacin	23.32	32	68	3.88
Erythromycin	18.55	26	74	4.52
Co-trimoxazole	22.32	38	62	3.98
Oxytetracycline	9.05	10	90	6.52

Health Organization also stated that AMR is the one of the major health risk in the 21st century and also mentioned as a major challenge to deal with public health (25). It has been estimated that across the globe approximately 0.7 million mortalities have been recorded due to infections with antibiotic-resistant bacteria and the numbers might reach 10 million by 2050 (26). Anthropogenic activities such as wastewater discharge, aquaculture, agriculture discharge, and hydraulic engineering are

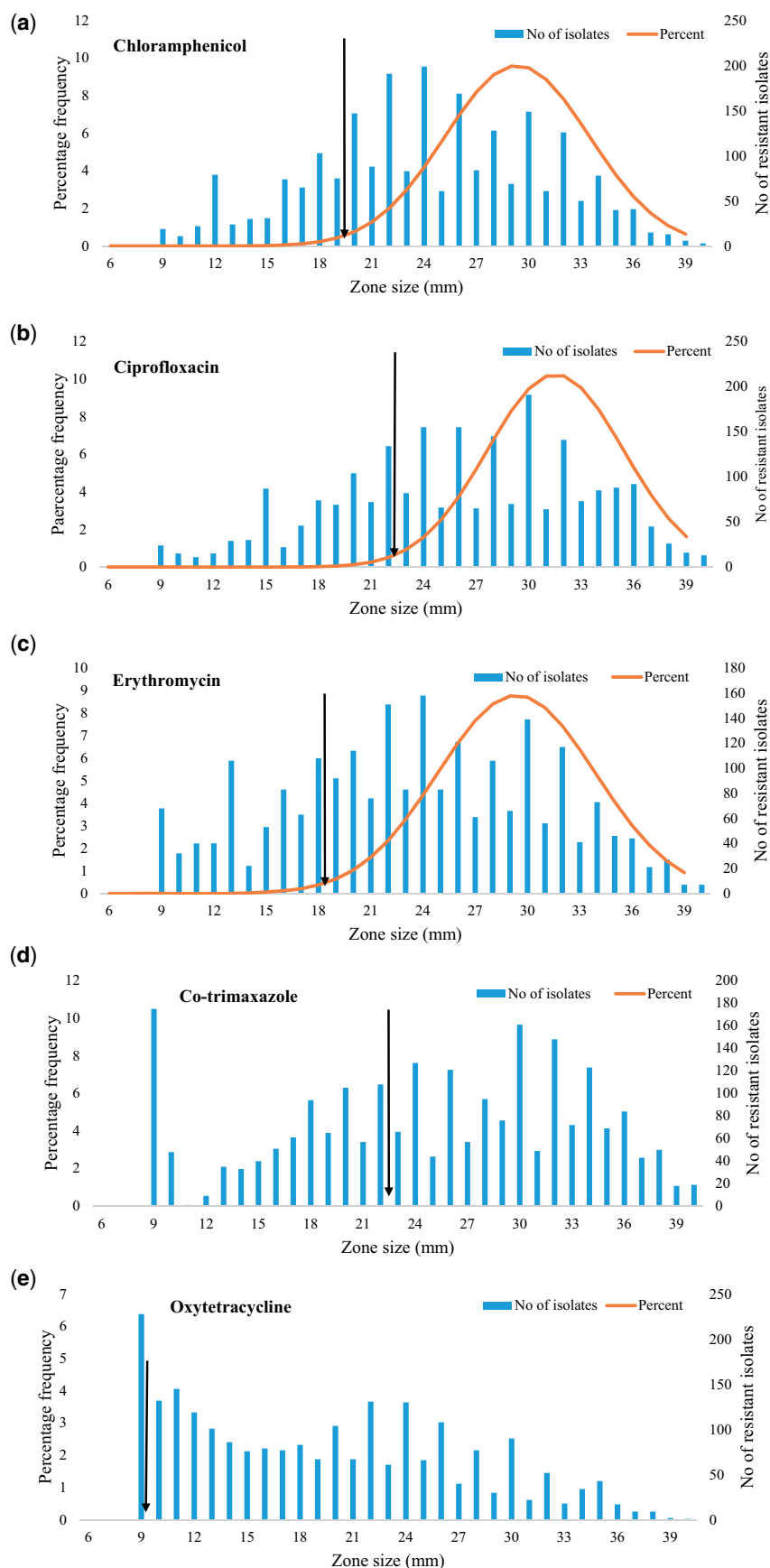


Figure 2. Distribution of inhibition zones formed for the antibiotic discs against heterotrophic bacteria: (a) chloramphenicol, (b) ciprofloxacin, (c) erythromycin, (d) co-trimoxazole, (e) oxytetracycline. The vertical dark line indicates calculated CO_{WT} value (epidemiological cutoff) and the continuous line indicates 8-point rolling means.

regarded as major sources of the acquisition and dissemination of AMR in aquatic environs (10). Aquatic water bodies are treated as chief reservoirs of antibiotic resistance genes (ARGs) and antibiotic-resistant bacteria and these are further transmitted through commensal and pathogenic bacterial populations to humans (27) and, as a consequence, make it difficult for medical practitioners to treat diseases and increase treatment costs that burdens patients (28). At present in India, the Coastal Aquaculture Authority (CAA) is the nodal agency which deals with the regulations regarding farming of *P. vannamei* in different regions of the country. The CAA advocates the use of antibiotic-free inputs in shrimp culture only and have provided a list of antibiotics that are prohibited for use in shrimp farming (<https://caa.gov.in/uploaded/doc/CAA-Guidelines-Antibiotic-Free%20Inputs.pdf>).

P. vannamei is widely cultured throughout the world, significantly contributing to the global economy (1). In order to reduce stress to the growing animals and achieve faster growth, farmers have been administering antibiotics in an uncontrollable manner that has resulted in the acquisition and spread of antimicrobial resistance in microbes prevailing throughout the food chain (29). There are several reports available on antibiotic use in shrimp aquaculture (30–32). In this study also, wide prevalence of antibiotic resistance was noticed in heterotrophic bacterial isolates. Highest resistance was noticed towards oxytetracycline by the bacterial isolates from location L6, as it was a preferred drug for application in aquaculture practices (33). The major factors for resistance to antibiotics in the rearing environment have been correlated to inappropriate usage of antibiotics, presence of residues, contamination of feed, seed (post-larvae) and water (9). Apart from these, antibiotic resistance also arises from several other factors including water run-off to the aquaculture systems, partial removal of water during harvesting time, and use of antibiotics for other livestock systems/food production systems. Antibiotic resistant genes carried by mobile genetic elements such as plasmids, integrons, and transposon played a significant role in the mediation of genes between heterotrophic bacterial populations in aquatic bacteria from shrimp aquaculture (34). Screening of antibiotic resistance in heterotrophic bacterial isolates is essential for understanding the resistance pattern in food production systems especially shrimp aquaculture.

Among the four groups of bacteria, Group2 (Gram-negative rods) were the most predominant followed by Group1 (Gram-positive rods), Group3 (Gram-positive cocci), and Group4 (Gram-negative coccobacilli). Among bacterial cultures, *Vibrio* spp. was the most dominant bacterial genera, similar to previous studies, in the shrimp culture environments with more drug resistance (6, 7, 9, 18, 42). This study also reports the prevalence of *Bacillus* spp. as the second most dominant bacterial species that are used as beneficial bacteria in shrimp culture systems for improving shrimp gut health and bioremediating pond water and sediment. Apart from *Vibrio* spp., the presence of opportunistic human pathogenic bacteria (*Burkholderia* spp., *Escherichia coli*, *Pseudomonas* spp., and *Staphylococcus* spp.) and fish pathogens (*Tenacibaculum* spp. and *Pseudoalteromonas* spp.) have also been observed in this study. The presence of *Acinetobacter*, *Bacillus*, *Burkholderia* spp., and some other bacteria isolated from shrimp culture ponds also have been previously reported (39–42). In addition, the prevalence of pathogenic bacteria such as *Acinetobacter* (43), *Pseudomonas* spp. (44), and *Escherichia coli* (45) has been reported from farm sediment samples. The aquatic environment in which these animals are reared likely has a role in the development and dissemination of AMR (29) and aquaculture operations may also contribute to this process. Antibiotic-resistant bacteria which are commonly found in humans

can be discharged into the aquaculture environment from sources such as run-off from agricultural fields and from treated and untreated municipal sewage waste waters (9). Aquaculture rearing facilities might also act as reservoirs for antibiotic-resistant bacteria and the mobile genetic elements they carry are able to transfer these genes to humans (34). High levels of antibiotic-resistant bacteria have been reported in the shrimp farm environments (30) which has also been noticed in the present study. The presence of antibiotic resistance in pathogenic bacteria has been previously reported by several authors (6–9, 34, 42).

The determination of antimicrobial breakpoints in different laboratories may vary with regard to adopted protocols, chemical reagents, and technical personnel performing the test, which affect the outcome greatly (different antimicrobial breakpoints) for a particular bacterium. Hence, the antimicrobial breakpoints determined employing the standard protocols (CLSI) can be used widely, are well accepted and are also recognized universally. A normalized distribution of the disc diffusion data has been noticed for all the studied antibiotics. In several previous studies, NRI has been successfully applied in calculating CO_{WT} of bacteria, especially pathogenic isolates (16, 18, 19, 35–37). Karatuna et al. (38) have studied ECOFF values for *Burkholderia pseudomallei* (n=361) against eight antibiotics and generated background data which may be useful in surveillance studies to set antimicrobial breakpoints for this pathogen. Similarly, ECOFF values for *Aeromonas* spp. isolated from aquaculture environments have been determined using NRI analysis and ECOFFinder from MIC data against eight antibiotics (18).

CO_{WT} for oxytetracycline was found to be ≥9 mm and 90% of the isolates were found to be the NWT phenotype, as this antibiotic has been used commonly in aquaculture and is also cheaply available (33). For other tested antibiotics, CO_{WT} has been found on the higher side and NWT phenotypes have been found on the lower side, especially towards ciprofloxacin, indicating that these antibiotics were used in lesser proportions compared to oxytetracycline. However, the situation is quite alarming since 60–80% of the heterotrophic isolates were found to be NWT. Similarly, *Aeromonas* spp. were found to be NWT strains towards florfenicol, enrofloxacin, and doxycycline (18). Therefore, due to lack of break-even points for heterotrophic bacteria, break-even points provided for pathogenic bacteria may be used for gauging resistance pattern in heterotrophic bacterial populations towards a particular antibiotic by means of ECOFF values.

Conclusions

In this study, heterotrophic bacterial populations were mainly classified into four groups based on morphotinctorial groups with *Vibrio* spp. (aquatic animal pathogens) being the dominant bacteria followed by *Bacillus* spp. (probiotic bacteria). CO_{WT} has been calculated for heterotrophs, since antimicrobial susceptibility studies for pathogenic bacteria require specific media, incubation conditions, media supplements, and appropriate methods. To study the antibiotic resistance patterns in heterotrophic bacteria requires general media and incubation at ambient temperature. AST studies for differentiation of sensitive and resistant in heterotrophic bacteria universally defined breakpoints for QC strains (*Staphylococcus aureus*, ATCC 29213 and *Escherichia coli*, ATCC 25922) may be employed for the same. However, the major drawback is the lack of a breakpoint for categorization of susceptible and resistant phenotypes for heterotrophic bacteria, whereas for pathogenic bacteria these are well defined by the CLSI. Therefore, AST studies for differentiation of sensitive and resistant in heterotrophic bacteria universally defined antimicrobial breakpoints for QC strains may be employed for the

same. ECOFF determination from disc diffusion data uses a low-cost protocol compared to MIC determination, which may be readily applied for studying antimicrobial breakpoints (CO_{WT}) of bacterial pathogens in low- and middle-income nations and can be easily applied in laboratories located in endemic regions (38). This study concluded NRI analysis for interpretation of antibiotic disc diffusion data from heterotrophic bacteria can be successfully applied for calculating ECOFF values as a whole. This is the first report on the calculation of CO_{WT} of heterotrophic bacteria prevalent in shrimp production systems.

CRediT Author Statement

R.K.N. Development of research concept, collection of shrimp farm samples, performing the experiments, drafting of original manuscript.

S.K.P. Development of research concept, and results interpretation.

M.R.B. Designing of experiments, collection of shrimp farm samples and editing of manuscript.

P.P.K. Arranged facilities in field collection and laboratory analysis, editing of manuscript and statistical analysis of data.

R.R.P. Development of research concept and research supervision.

A.K. Data interpretation and calculation of ECOFF data.

U.D. Data interpretation and calculation of ECOFF data.

P.K. Identification of heterotrophic bacteria.

M.P.M. Development of research concept, design of sample collection, acquisition of fund, editing of manuscript and overall supervision.

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Conflict of Interest

There is no conflict of interest in publishing this research manuscript.

Supplemental Information

Supplemental information is available on the J. AOAC Int. website.

Data Availability

The complete data set used for interpretation is available from the corresponding author.

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